



Electrical and Computer Engineering
University of Thessaly

ECE214 - Introduction to Electronics

SPICE-Based DC and AC Analysis of CMOS Common-Source Amplifiers

Spring Semester — 2021-2022

Assignment 5
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Exercise 8.1

The amplifier whose DC analysis is taking place in the current exercise is illustrated in figure 1. Beyond the circuit picture the current section is divided into subsections each containing the answers of the task.

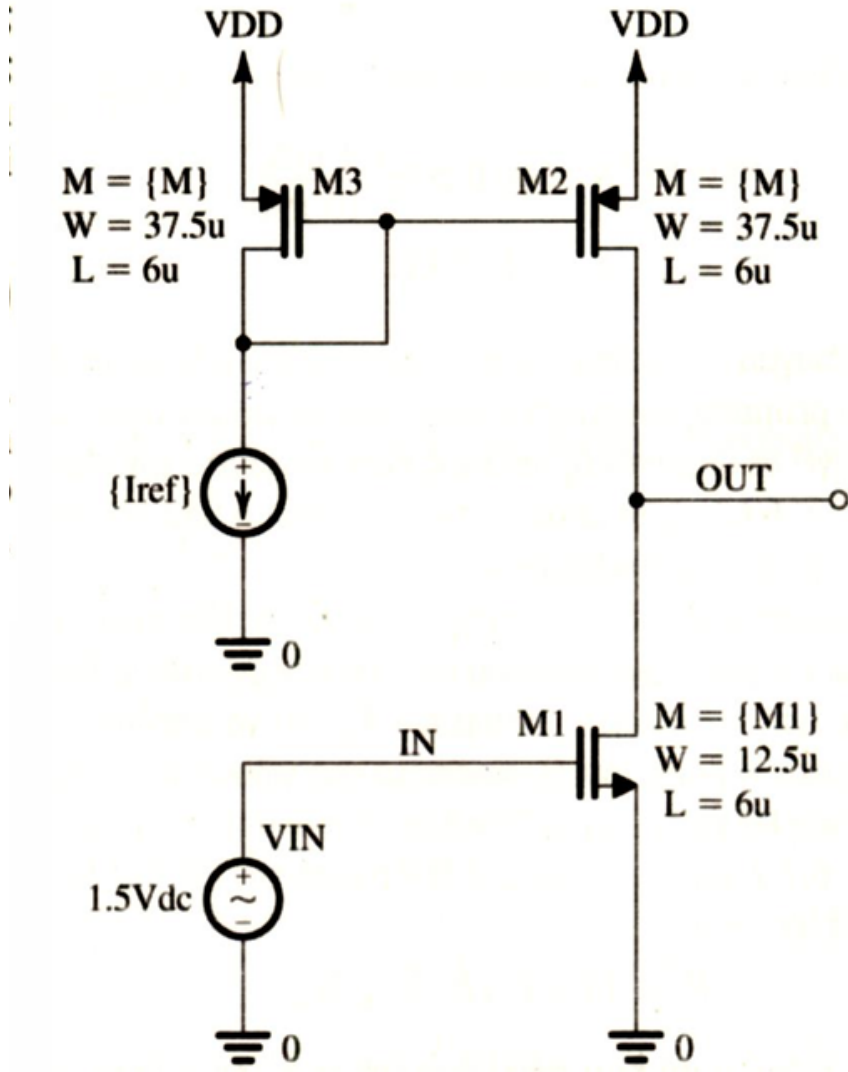


Figure 1: Amplifier circuit implementation

(a)

In order to calculate the dimensions of transistors M_1 , M_2 , and M_3 , the overdrive voltages are assumed to be $V_{OV1} = 0.1$ V and $V_{OV2} = V_{OV3} = 0.2$ V. Additionally, it is given that $k'_n = 2.5 \times 10^{-4}$ A/V², $k'_p = 5.19 \times 10^{-5}$ A/V², $I_D = I_{ref} = 100$ μ A, and $L = 0.25$ μ m.

For the NMOS transistor M_1 , the drain current is given by:

$$I_D = \frac{1}{2} k'_n \frac{W_n}{L} V_{OV1}^2 \quad (1)$$

Therefore, it follows that:

$$W_1 = W_n = 20 \mu\text{m} \quad (2)$$

Similarly, for the PMOS transistors M_2 and M_3 , the drain current is given by:

$$I_D = \frac{1}{2} k'_p \frac{W_p}{L} V_{OV2}^2 \quad (3)$$

Therefore, it follows that:

$$W_2 = W_3 = W_p = 24 \mu\text{m} \quad (4)$$

(b)

By analyzing the circuit, the small-signal voltage gain (G_v) is derived using the following mathematical expression:

$$G_v = -\frac{2}{V_{OV1}} \cdot \frac{V_{An} \cdot V_{Ap}}{V_{An} + V_{Ap}} \quad (5)$$

The Early voltages for both the NMOS and PMOS transistors are determined by the channel-length modulation parameter (λ) according to the formula:

$$V_A = \frac{1}{\lambda} \quad (6)$$

Given the channel-length modulation coefficients:

$$\begin{aligned} \lambda_n &= 0.11 \text{ V}^{-1} \\ \lambda_p &= 0.09 \text{ V}^{-1} \end{aligned}$$

The respective Early voltages are calculated as follows:

$$V_{An} = \frac{1}{\lambda_n} = \frac{1}{0.11 \text{ V}^{-1}} \approx 9.09 \text{ V} \quad (7)$$

$$V_{Ap} = \frac{1}{\lambda_p} = \frac{1}{0.09 \text{ V}^{-1}} \approx 11.11 \text{ V} \quad (8)$$

Substituting these values into the gain equation yields the final voltage gain of the amplifier:

$$G_v = -100 \text{ V/V} \quad (9)$$

(c)

The voltage range where transistors M_1 and M_2 remain in the saturation region, and consequently, the amplifier operates in the linear region, is defined as follows:

$$V_{OV1} \leq V_{out} \leq V_{DD} - V_{OV2} \quad (10)$$

Which yields:

$$0.1 \text{ V} \leq V_{out} \leq 2.3 \text{ V} \quad (11)$$

(d)

With the usage of LTSpice we conduct a DC analysis in order to simulate the behaviour of the amplifier when an steadily increasing voltage (V_{in}) is applied to its input. The output voltage response is illustrated in figure 2.

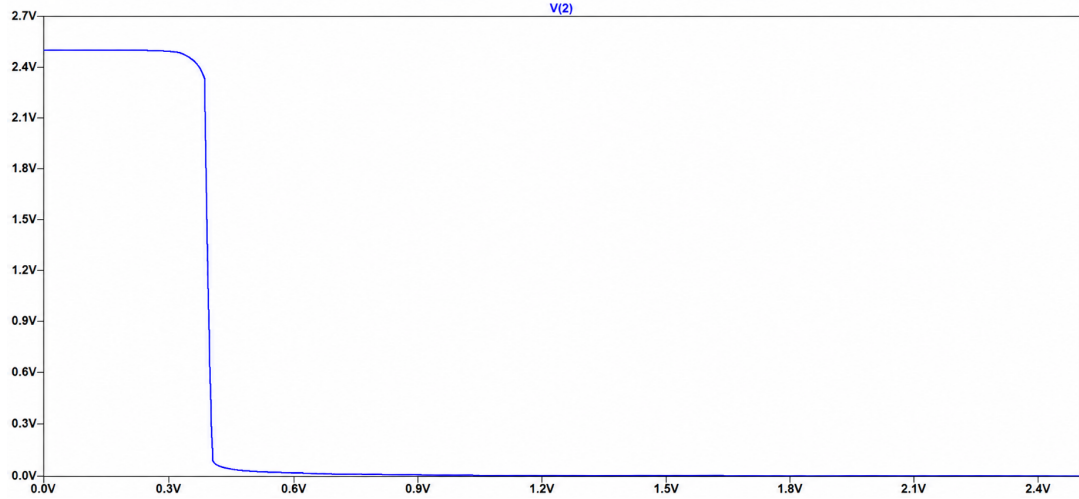


Figure 2: LTSpice Transient Response of the Amplifier

It is observed that the linear operating region of the amplifier is bounded approximately by the input voltages $V_{in} = 0.39$ V and $V_{in} = 0.409$ V. The corresponding output voltages are $V_{out} = 2.27$ V and $V_{out} = 0.13$ V, which are in close agreement with the theoretical values. Consequently, no parameters need to be altered, and the amplitudes do not require recalculation.

(e)

Using the tools of LTSpice we obtain a derivative function of the transient response characteristic (3). Based on the results mentioned above, the voltage gain (G_u), which represents the slope of V_{out} within the linear region, is approximately -114.74 . This value is acceptably close to the one obtained by hand analysis.

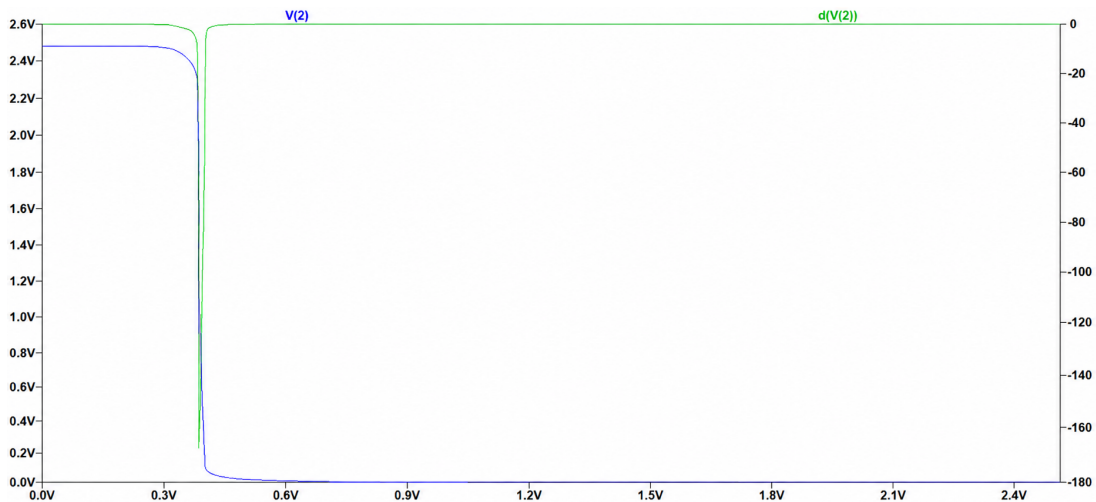


Figure 3: $\frac{dV_{out}}{dV_{in}}$ function of Amplifier

(f)

From the figure 2, it is observed that for a biasing voltage of $V_{in} = 0.399$ V, the DC output biasing voltage is $V_{out} = 1.03$ V. This input biasing voltage, which yields an output approximately at the midpoint of the amplifier's linear region, provides the maximum voltage gain as well as the largest available output signal swing margin.

(g)

By setting the input voltage at $V_{in} = 0.399$ V we conduct the following .op analysis (figure 4).

```

--- Operating Point ---

V(1) :          0.399          voltage
V(2) :          1.04033        voltage
Id(M3) :         -0.0001        device_current
Id(M2) :         -0.000102764    device_current
Id(M1) :          0.00010279     device_current

```

Figure 4: .op Analysis of Amplifier

The results obtained from the .op analysis of the amplifier verify the values of V_{out} , I_{d1} , I_{d2} , I_{d3} given as a clue when the amplifier operates in maximum gain.

(h)

A sinusoidal wave with an amplitude of 0.008 V and a frequency of 1 kHz is applied now in the amplifier as an input, which is vertically offset from zero by the biasing voltage of 0.399 V (first waveform of figure 5). In the second waveform of the figure 5 the output is illustrated where except the aptitude difference exhibits a phase shift of π rad compared to the first one.

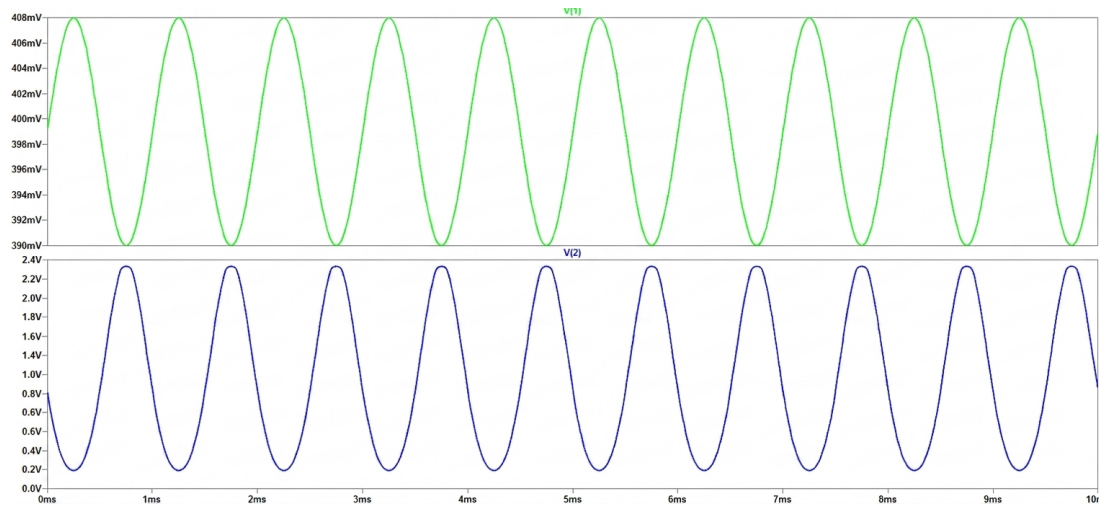


Figure 5: Input Sine Voltage and Output response of the Amplifier

By measuring the peak-to-peak voltage at the input $V_{pp,in} = 16 \text{ mV}$ and at the output $V_{pp,out} = 1.94 \text{ V}$, and calculating their ratio, we obtain:

$$V_{pp,out} \approx 120 \cdot V_{pp,in} \quad (12)$$

Consequently, the output voltage is approximately 120 times larger than the input voltage. Therefore, the voltage gain (A_v) is approximately 120, which is very close in absolute value to the two gain values previously calculated.

The maximum amplitude of the sinusoidal input signal for which we achieve the desired amplification without output distortion (i.e., clipping) is equal to 0.008 V . For larger amplitudes, the output waveform is clipped, and its sinusoidal shape becomes distorted.

Exercise 9.1

In the current exercise, an AC analysis is conducted for the common-source amplifier, whose circuit implementation is illustrated in Figure 6.

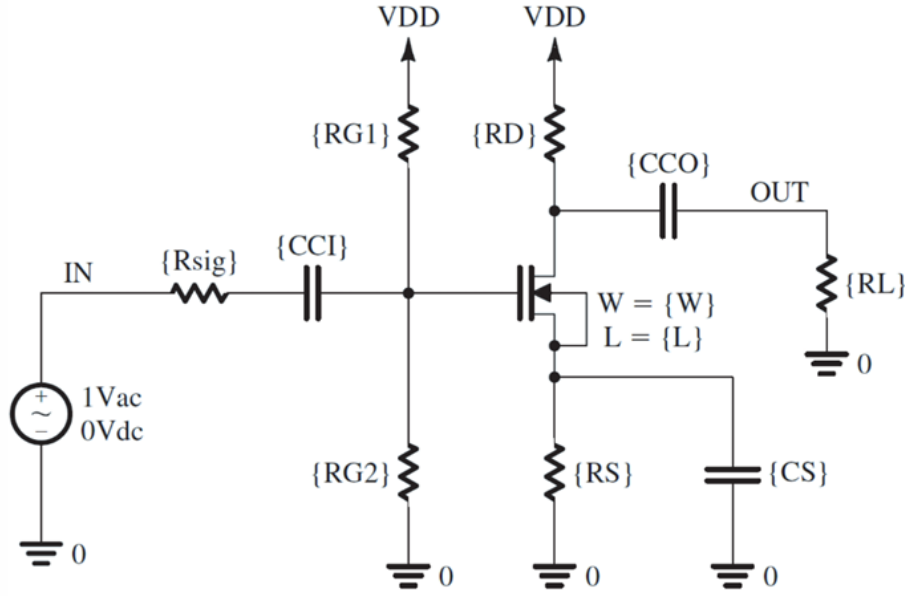


Figure 6: Common-Source Amplifier with a Resistive Load Circuit Implementation

(a)

Initially, the given operating parameters and design specifications are defined as follows:

$$\begin{aligned} V_{DD} &= 2.5 \text{ V} \\ P &= 1.5 \text{ mW} \\ |A_v| &= 10 \text{ V/V} \\ L &= 0.25 \mu\text{m} \end{aligned}$$

The drain bias current I_D is calculated from the power consumption equation:

$$I_D = \frac{P}{V_{DD}} = \frac{1.5 \text{ mW}}{2.5 \text{ V}} = 0.6 \text{ mA} \quad (13)$$

The overdrive voltage is chosen arbitrarily as $V_{OV} = 0.06 \text{ V}$. Additionally, the channel-length modulation parameter is $\lambda = 0.11 \text{ V}^{-1}$ (for the NMOS transistor), and the process transconductance parameter is $k'_n = 2.5 \times 10^{-4} \text{ A/V}^2$.

The drain-to-source voltage is established at $V_{DS} = \frac{V_{DD}}{3} \approx 0.833 \text{ V}$. Substituting these values into the transistor drain-current equation for the saturation region:

$$\frac{W}{L} = \frac{I_D}{\frac{1}{2}k'_n V_{OV}^2 (1 + \lambda V_{DS})} \quad (14)$$

By evaluating the expression with the given parameters, the transistor aspect ratio is determined, yielding a gate width of $W = 305 \mu\text{m}$ for the specified channel length L .

Furthermore, to determine the required drain resistance R_d , the small-signal transconductance g_m and the output resistance r_o are evaluated:

$$g_m = \frac{2I_D}{|V_{OV}|} = \frac{2 \cdot 0.6 \text{ mA}}{0.06 \text{ V}} = 0.02 \text{ A/V} \quad (15)$$

$$r_o = \frac{|V_A|}{I_D} = \frac{1}{D} = 15.15 \cdot 10^3 = 15.15k \quad (16)$$

The drain resistance R_D is then determined by solving the small-signal voltage gain equation:

$$|A_V| = g_m(R_D \parallel R_L \parallel r_o) \quad (17)$$

Yielding a calculated value of $R_D = 522.47 \Omega$.

Next, the output voltage is evaluated as follows:

$$V_o = V_{DD} - I_D R_D = 2.17 \text{ V} \quad (18)$$

Using the previously established condition $V_{DS} = \frac{V_{DD}}{3}$, the source resistance R_S can be calculated from the voltage drop equation across the source branch ($V_D - V_S = V_{DS} \Rightarrow V_S = V_D - \frac{V_{DD}}{3}$):

$$R_S = \frac{V_D - \frac{V_{DD}}{3}}{I_D} = 2255.3 \Omega \quad (19)$$

Finally, to determine the biasing resistors R_{g1} and R_{g2} , the gate voltage V_G must be found. Using the loop equation around the gate-source junction:

$$V_G = I_D R_S + V_{OV} + V_t = 1.83 \text{ V} \quad (20)$$

Applying the standard voltage divider formula at the gate network, we have:

$$V_G = V_{DD} \frac{R_{g2}}{R_{g1} + R_{g2}} \quad (21)$$

Substituting $V_G = 1.83 \text{ V}$ and $V_{DD} = 2.5 \text{ V}$ into the divider equation yields the ratio $R_{g2} \approx 2.73 \cdot R_{g1}$ (approximated as $R_{g2} = 3R_{g1}$ according to the design criteria). Assuming a high input impedance choice of $R_{g1} = 1 \text{ M}\Omega$, the second biasing resistor is chosen as:

$$R_{g2} = 3 \text{ M}\Omega \quad (22)$$

(b)

By conducting an operating point (.op) analysis using the resistance values $R_{g1} = 1 \text{ M}\Omega$ and $R_{g2} = 3 \text{ M}\Omega$, the simulation yields the results illustrated in Figure 7. A careful investigation of these results reveals that the DC output voltage $V(5)$ is equal to 2.14 V, which deviates slightly from the analytically calculated value. Similarly, the drain current of the NMOS transistor $I_d(M_1)$ also exhibits a minor discrepancy compared to the theoretical value of 0.6 mA obtained from the hand calculations.

```

      --- Operating Point ---

V(100) :      2.5          voltage
V(4) :      1.51364       voltage
V(5) :      2.14934       voltage
V(2) :      1.875e-013    voltage
V(3) :      1.875         voltage
V(6) :      1.07467e-012  voltage
V(1) :      0             voltage
Id(M_m1) :    0.00067115  device_current
Ig(M_m1) :    0             device_current
Ib(M_m1) :   -6.45703e-013 device_current
Is(M_m1) :   -0.00067115  device_current
I(C_co) :    2.14934e-017  device_current
I(C_ci) :   -1.875e-017    device_current
I(C_s) :   -1.51364e-017  device_current
I(R_sig) :   -1.875e-017    device_current
I(R_l) :    2.14934e-017    device_current
I(R_g2) :   -6.25e-007     device_current
I(R_g1) :   -6.25e-007     device_current
I(R_d) :   -0.000671149    device_current
I(R_s) :   -0.000671149    device_current
I(V_in) :    1.875e-017    device_current
I(V_dd) :   -0.000671774    device_current

```

Figure 7: Output results from the .op analysis

Therefore, the biasing resistors R_{g1} and R_{g2} are adjusted. By setting $R_{g1} = 12.8 \text{ M}\Omega$ and $R_{g2} = 30 \text{ M}\Omega$, the simulation results obtained from the operating point (.op) analysis are validated, matching the analytical design specifications (figure 8).

```

--- Operating Point ---

V(100) :      2.5          voltage
V(4) :      1.39452       voltage
V(5) :      2.17694       voltage
V(2) :      1.75234e-013  voltage
V(3) :      1.75234       voltage
V(6) :      1.08847e-012  voltage
V(1) :      0            voltage
Id(M_m1) :    0.000618332  device_current
Ig(M_m1) :    0            device_current
Ib(M_m1) :   -7.92419e-013 device_current
Is(M_m1) :   -0.000618332  device_current
I(C_co) :     2.17694e-017  device_current
I(C_ci) :    -1.75234e-017  device_current
I(C_s) :     -1.39452e-017  device_current
I(R_sig) :    -1.75234e-017  device_current
I(R_l) :     2.17694e-017  device_current
I(R_g2) :    -5.84112e-008  device_current
I(R_g1) :    -5.84112e-008  device_current
I(R_d) :     -0.000618331  device_current
I(R_s) :     -0.000618331  device_current
I(V_in) :     1.75234e-017  device_current
I(V_dd) :    -0.000618389  device_current

```

Figure 8: Output results from the .op analysis

(c)

The transient voltage response of the amplifier to a sinusoidal input, at the points before and after the CCO capacitor, is illustrated in Figure 9. The up subplot represents the output behaviour, regarding the point before the CCO capacitor while the down subplot describes the circuit behaviour in the point after the same capacitor.

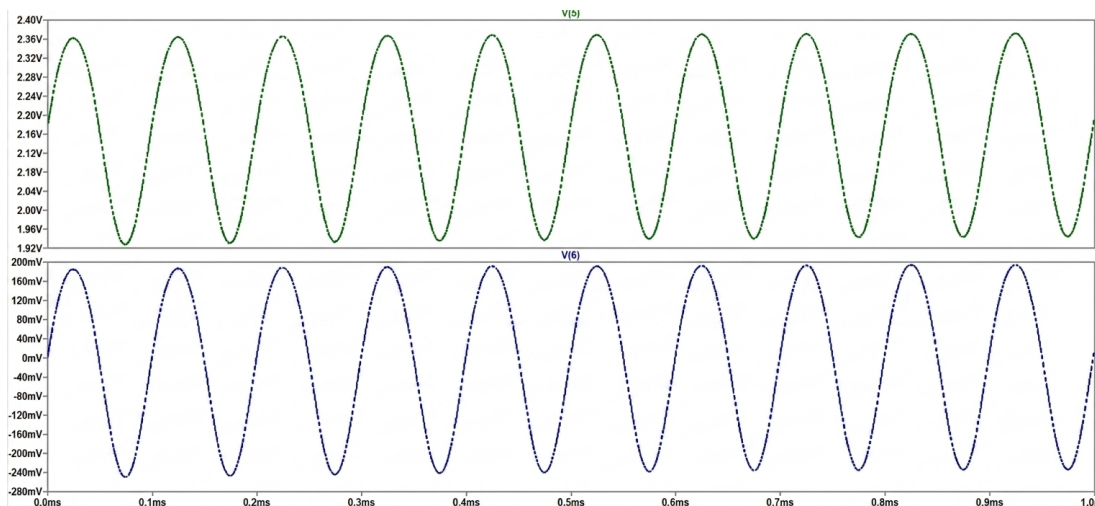


Figure 9: Voltage Transient Response before and after the Capacitor CCO

The output signals before and after the CCO capacitor exhibit the same phase, but they are biased at different DC voltage levels. Specifically, before the capacitor, the output oscillates

between 1.94 V and 2.36 V. After the capacitor, it shifts to oscillate between -238 mV and 191 mV.

This behavior demonstrates that the capacitor functions as a coupling (or DC-blocking) capacitor. It successfully filters out the DC component of the signal and shifts it around the reference voltage (near 0), while allowing the AC signal to pass through without any significant attenuation of its amplitude.

A clearer representation of the common-source amplifier's functionality is provided in Figure 10, which contrasts the input and output signals to illustrate the significant amplification of the input sine wave.

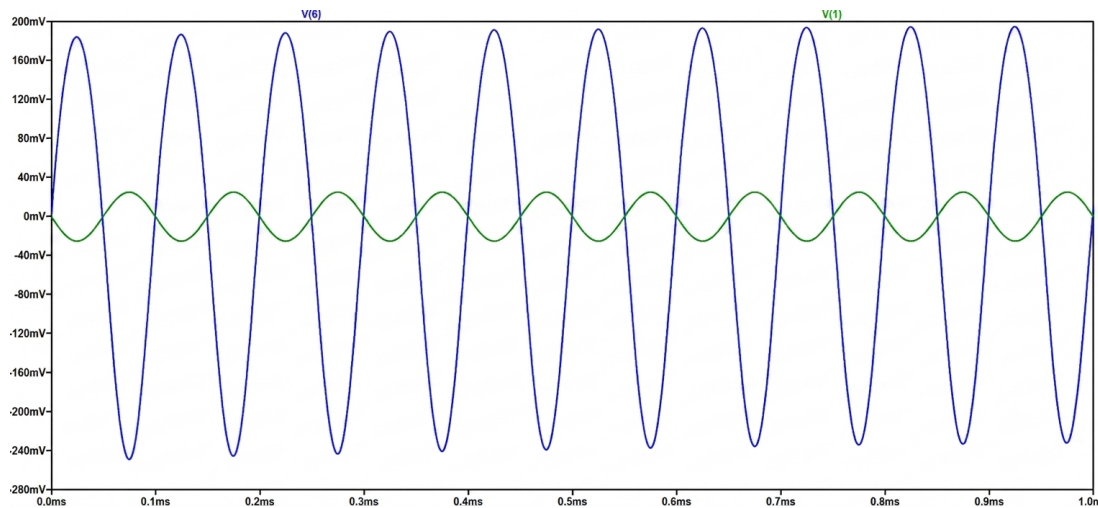


Figure 10: Input and Output Signals of the Amplifier

By gradually increasing the input signal amplitude, it is observed that the output waveform begins to distort, losing its sinusoidal shape and experiencing clipping once the input amplitude exceeds 0.06 V. Under these conditions, high-amplitude input signals are amplified uniformly, meaning that a constant output voltage is produced for different input voltage levels. Consequently, the voltage gain of the amplifier decreases due to the reduction of the peak-to-peak ratio ($V_{pp,out}/V_{pp,in}$). This phenomenon occurs because the amplifier ceases to operate within its linear region, leading to a sub-optimal gain. As illustrated in figure 11, the output waveform is starting to distort.

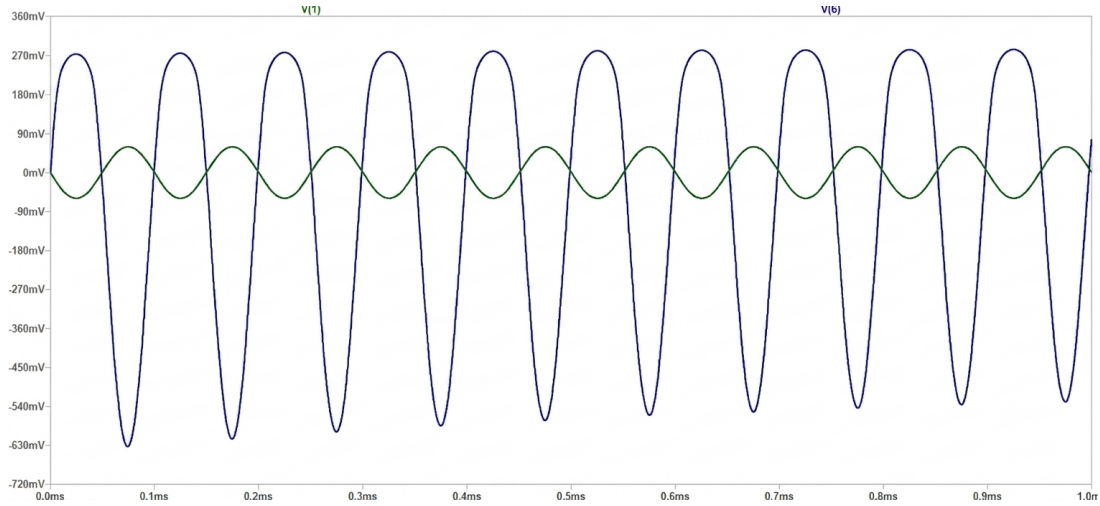


Figure 11: Input and Output Signals of the Amplifier for a distorted output

The output signal maintains its waveform shape as long as the input sine wave amplitude remains within the range of -60 mV to 60 mV, which causes the output amplitude to scale between -630 mV and 270 mV.

(d)

For a capacitor value of $C_s = 10 \mu\text{F}$, the output voltage frequency response of the amplifier is illustrated in Figure 12. According to Figure 12, it is evident that the midband voltage gain, where the output voltage stabilizes, is equal to 18.3 dB. From this observation, the 3 dB cutoff frequencies (f_L and f_H) can be determined. These values are derived from the frequencies where the voltage gain drops by 3 dB below its midband value, corresponding to a level of 15.3 dB.

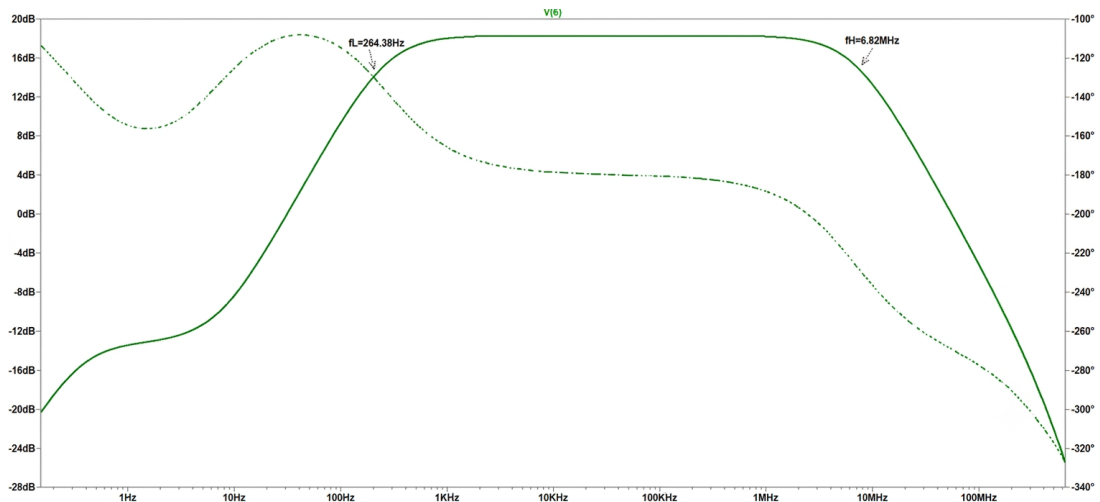


Figure 12: Output Voltage Response of the Amplifier on dB scale

Analysing the output voltage behaviour we conclude that the 3 dB frequencies are estimated at $f_L = 264.38$ Hz and $f_H = 6.82$ MHz.

By employing the following formula, the midband voltage gain is calculated to be $|A_u| = 8.31$ V/V:

$$|A_u|_{dB} = 20 \log |A_u| \quad (23)$$

This value is in close agreement with the experimental data of the exercise (-0 V/V). Finally, the 3 dB bandwidth (BW) is determined as follows:

$$BW = f_H - f_L = 6.95 \times 10^6 \text{ Hz} - 262.44 \text{ Hz} = 6,949,737.56 \text{ Hz} \approx 6.949 \text{ MHz} \quad (24)$$

It is observed that the amplifier attenuates the gain at low frequencies. Similarly, for $C_s = 0$, the extracted small-signal parameters are:

$$\begin{aligned} f_L &= 0.317 \text{ Hz} \\ f_H &= 185.11 \text{ Hz} \\ A_u &= 0.21 \text{ V/V} \end{aligned}$$

The corresponding 3 dB bandwidth (BW) is determined as follows:

$$BW = f_H - f_L = 74.4 \text{ MHz} - 317.9 \text{ mHz} \approx 74.4 \text{ MHz} \quad (25)$$

The figure 13 represents the output response of the amplifier for both case ($C_s = 0, 10F$). Examining the figure we observe that when the bypass capacitor C_s is removed, no attenuation occurs at low and high frequencies. However, as is evident, the voltage gain is significantly reduced (expressed in negative dB), which is attributed to the presence of the source resistance R_s . Specifically, the midband voltage gain (A_M) is degraded by a factor approximately equal to $1 + g_m R_s$:

$$1 + g_m \cdot R_s = 1 + 0.02 \text{ A/V} \cdot 2255.3 \Omega = 46.106 \quad (26)$$

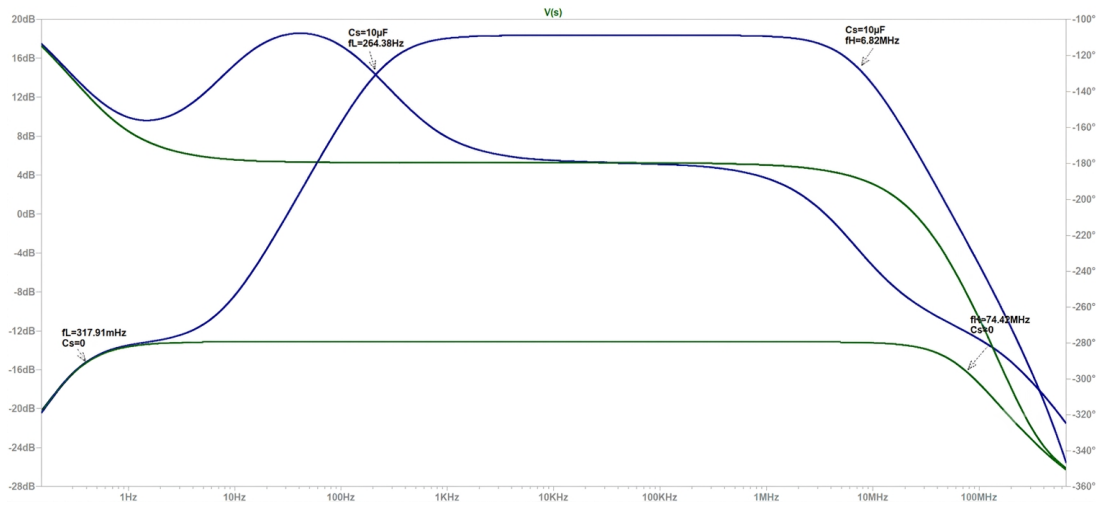


Figure 13: Output Voltage Response of the Amplifier on dB scale for case A ($C_s = 0$) and case B ($C_s = 10 \mu\text{F}$)

E)

Following a substantial number of iterations, the obtained results demonstrate high validity. All underlying numerical computations and analysis were performed utilizing MATLAB.